

Probability of Default Scorecard

WoE-Based Retail PD Prototype with Out-of-Time Backtesting

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1 What this project does

This is a retail Probability of Default (PD) scorecard prototype built end-to-end on a synthetic loan portfolio. The model follows the methodology used by European banks for IRB scorecards: Weight-of-Evidence (WoE) binning, Information-Value-based feature selection, regularized logistic regression on WoE-encoded features, intercept calibration to the long-run-average default rate, mapping to a rating master scale, and out-of-time backtesting.

It is a portfolio demonstration, not a regulator-approved model. The goal is to show that I can implement the structural components of an IRB scorecard end-to-end and reason about how they connect to Basel and IFRS 9 requirements. A production-grade version would need real bureau data, a defined default trigger, independent validation, and the full model governance infrastructure described in the limitations section.

2 Data

A synthetic portfolio of 250,000 unsecured retail loans, 25,000 per origination vintage across ten half-year cohorts (2019H1 through 2023H2). The dataset spans five years, which matches the Basel minimum observation period for retail IRB PD estimation. Each loan has an 18-month performance window and a realized default outcome.

The data-generating process embeds a logit model where bureau attributes (score, prior bankruptcies, worst delinquency status, inquiries, utilization) are the strong predictors, application attributes (DTI, income, employment) are moderate predictors, and demographics (age, loan purpose) are weak predictors. Vintage-level fixed effects produce cohort-to-cohort variation in the realized default rate, mimicking the macroeconomic path from the pre-COVID benign cohorts, through the 2020 stress (masked by forbearance) and the 2021 recovery, and into the 2022–2023 rate-hike pressure.

Vintage	Loans	Defaults	Default rate
2019H1	25 000	2 513	10.05%
2019H2	25 000	2 556	10.22%
2020H1	25 000	2 750	11.00%
2020H2	25 000	2 939	11.76%
2021H1	25 000	2 866	11.46%
2021H2	25 000	2 764	11.06%
2022H1	25 000	3 041	12.16%
2022H2	25 000	3 141	12.56%
2023H1	25 000	3 399	13.60%
2023H2	25 000	3 605	14.42%
Total	250 000	29 574	11.83%

Table 1: Portfolio composition by origination vintage. Realized default rates trend from 10.05% in the pre-COVID cohort to 14.42% in the 2023H2 stress cohort.

2.1 Cohort split

The split is chronological by vintage, as is standard in IRB modelling, because covariate drift across cohorts is the dominant failure mode in production:

- **Training:** 2019H1 through 2021H2 (six cohorts, 150 000 loans).

- **Validation:** 2022H1 + 2022H2 (two cohorts, 50 000 loans), in-time.
- **Out-of-time:** 2023H1 + 2023H2 (two cohorts, 50 000 loans), the worst-performing cohorts in the panel, used as a stress test.

3 Methodology

3.1 Weight of Evidence binning

Each feature is transformed from raw value to a WoE score that captures its monotonic relationship with default. For a bin b of feature X , WoE is defined as

$$\text{WoE}_b = \log \frac{\Pr(X \in b \mid y = 0)}{\Pr(X \in b \mid y = 1)}$$

positive when non-defaulters are over-represented in the bin, negative otherwise.

Numeric features are binned into deciles. Low-cardinality numeric features (counts of bankruptcies, public records, missingness flags) use their distinct values directly as bins. Categorical features (home ownership, loan purpose) are encoded with WoE per category. All encoders are fit on training data only and applied unchanged to validation and OOT cohorts.

3.2 Information Value and feature selection

For each feature, the Information Value summarizes how much the binned distribution of X differs between defaulters and non-defaulters:

$$\text{IV} = \sum_b (\Pr(X \in b \mid y = 0) - \Pr(X \in b \mid y = 1)) \cdot \text{WoE}_b$$

The industry convention is to drop features with IV below 0.02, on the basis that anything weaker is essentially uninformative and only adds noise to the regression.

Feature	IV
bureau_score	0.601
interest_rate_pct	0.512
utilization_pct	0.476
worst_status_24m	0.331
delinquencies_2y	0.283
months_since_last_delinquency_missing	0.263
months_since_last_delinquency	0.140
prior_bankruptcies	0.105
inquiries_6m	0.097
public_records	0.026
<i>Below 0.02 threshold (dropped)</i>	<i>16 features</i>

Table 2: Information Value per feature, kept set above the 0.02 threshold. Bureau-derived attributes dominate as expected.

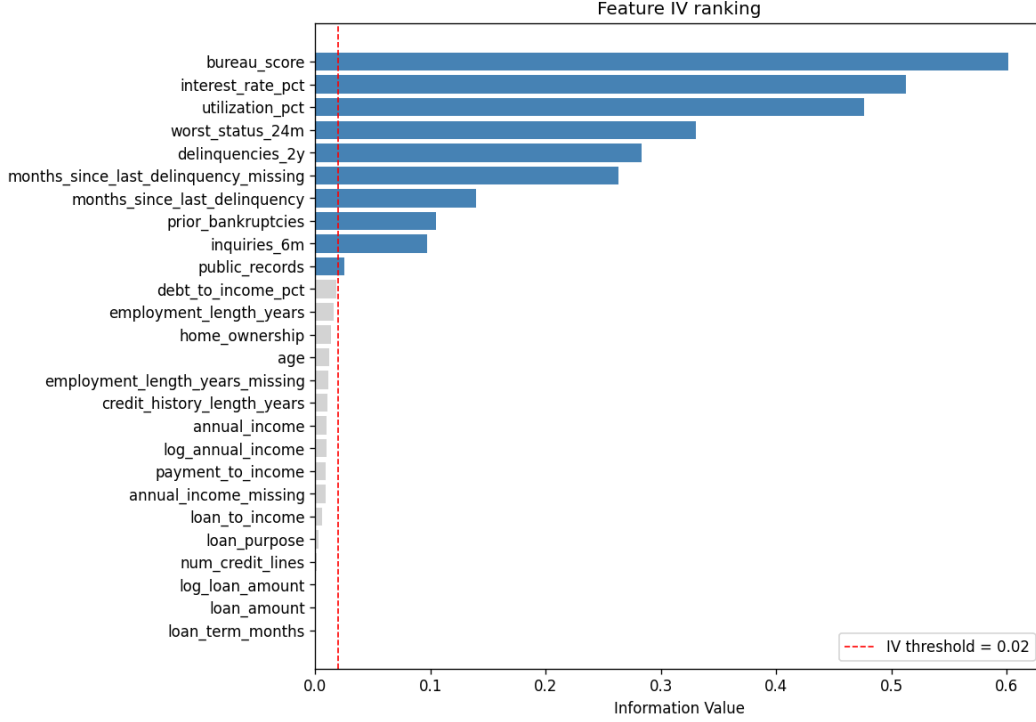


Figure 1: Information Value ranking. Bars above the red threshold are kept; below are dropped.

The final model uses ten features. All are bureau-derived or strongly bureau-influenced (interest rate, set by the lender based on the applicant’s risk profile). Application-only attributes (DTI, employment length, income, age) all fall below the IV threshold once the bureau-based features are in.

3.3 Logistic regression on WoE features

The model is L2-regularized logistic regression with $C = 1.0$:

$$\log \frac{p_i}{1 - p_i} = \beta_0 + \sum_j \beta_j \cdot \text{WoE}_j(x_{ij})$$

fitted by maximum likelihood on the training cohort.

3.4 Calibration to long-run-average PD

The raw logistic output is rank-discriminative but the level can drift when training and prediction populations differ. The standard IRB fix is an additive intercept correction:

$$\widehat{\text{PD}}_i = \frac{1}{1 + \exp(-(\text{logit}(\hat{p}_i^{\text{aw}}) + c))}$$

where c is chosen so that the mean of $\widehat{\text{PD}}$ on the training set equals the realized default rate across the six training cohorts. Because the training panel spans pre-COVID, COVID, and recovery, the resulting calibration target approximates a through-the-cycle average rather than any single vintage.

3.5 Master rating scale

PDs are mapped to a nine-grade scale (A1 to C3), constructed as piecewise cutpoints in PD space. The grades are the unit of currency for the rest of the bank: loan pricing, capital calculation, and

IFRS 9 staging all run on grades, not on raw PD scores. The grade boundaries are 0.5%, 1%, 2.5%, 5%, 8%, 12%, 18%, 28%, and 100%.

4 Results

4.1 Discrimination

Cohort	AUC	Gini	KS	Log-loss	Brier
TRAIN (150k, 2019–2021)	0.7778	0.5557	0.4126	0.2915	0.0846
VAL 2022 (50k)	0.7699	0.5397	0.4038	0.3210	0.0951
OOT 2023 (50k)	0.7693	0.5386	0.3991	0.3492	0.1048

Table 3: Headline metrics across the chronological split. Gini sits in the upper half of the typical retail PD range (0.40–0.60 acceptable, 0.55+ strong) and drops only 1.7 Gini-points from train to out-of-time despite spanning a full macroeconomic cycle.

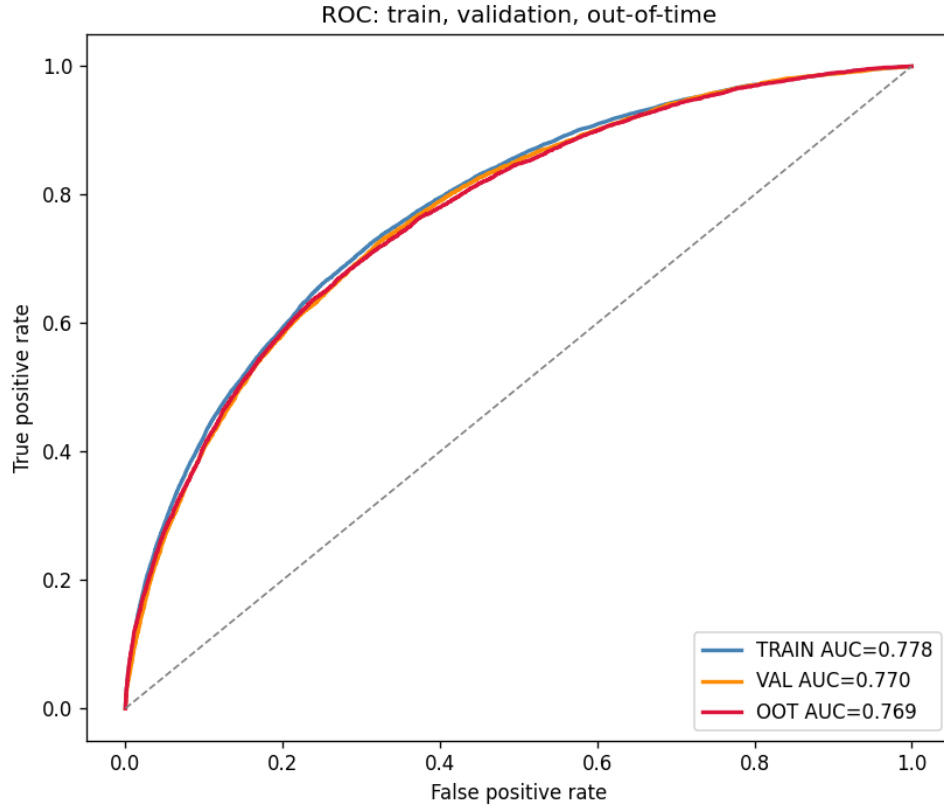


Figure 2: ROC curves across train, validation, and out-of-time. The three curves are nearly super-imposable.

4.2 Calibration per rating grade

Grade	N (OOT)	Pred PD	Observed DR	Share
A2	1 314	0.86%	0.91%	2.6%
A3	7 136	1.77%	2.59%	14.3%
B1	10 675	3.68%	5.56%	21.4%
B2	8 413	6.37%	9.14%	16.8%
B3	7 202	9.84%	12.77%	14.4%
C1	6 160	14.75%	18.86%	12.3%
C2	5 041	22.28%	28.47%	10.1%
C3	4 059	39.95%	47.50%	8.1%

Table 4: Predicted vs observed default rate per grade on the 2023 out-of-time cohort. The TTC-calibrated model under-predicts realized rates in the B and C grades by 2–7 percentage points, reflecting the macroeconomic stress that a TTC scorecard is not designed to track point-in-time.

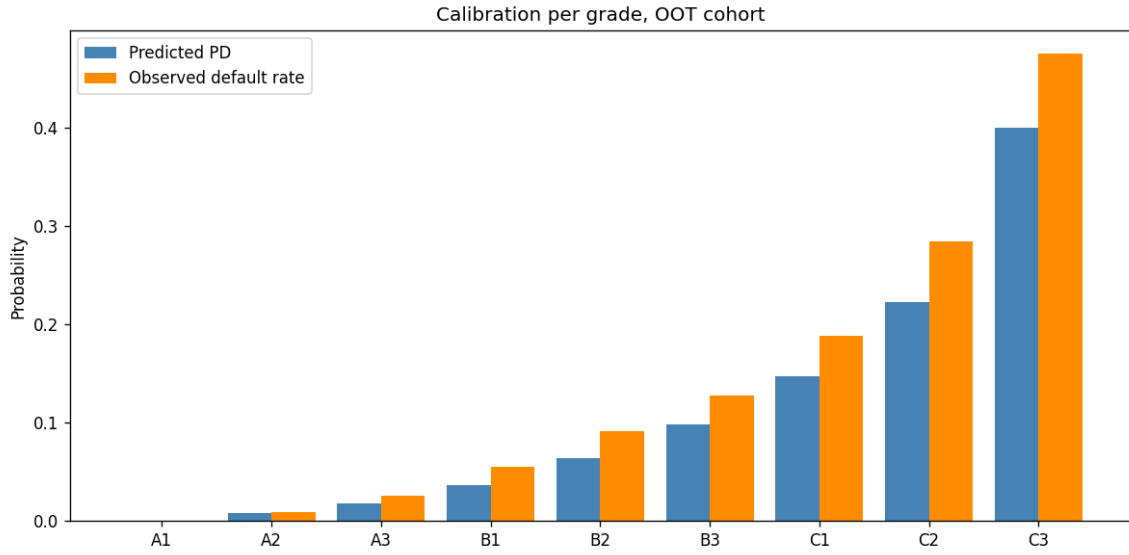


Figure 3: Predicted PD vs observed default rate per grade on the 2023 out-of-time cohort.

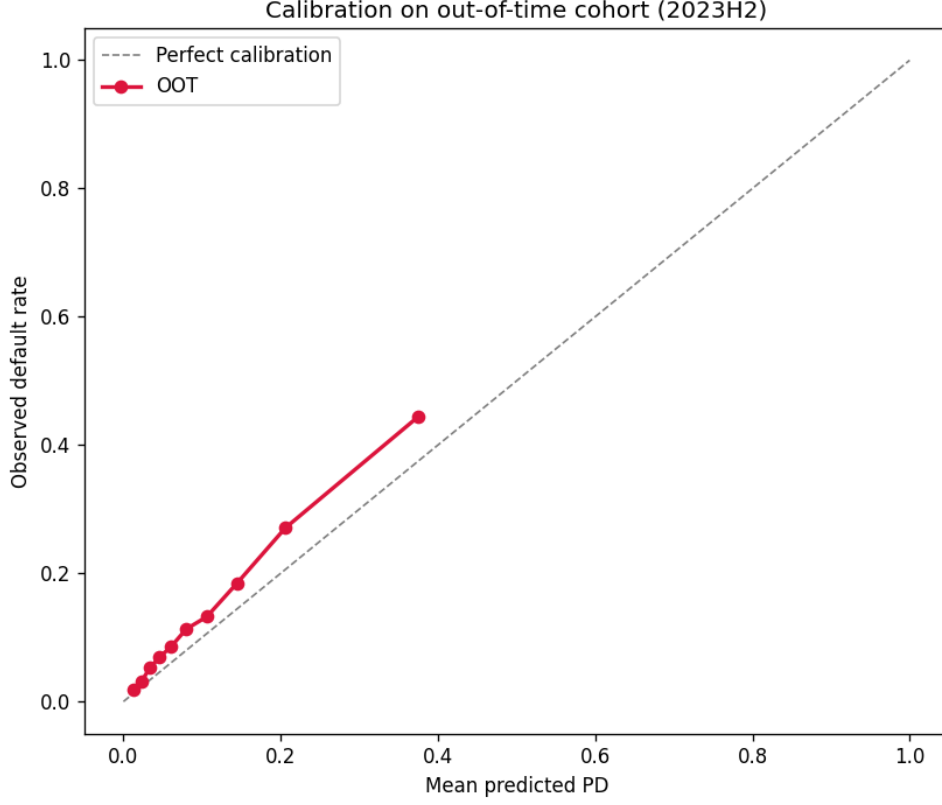


Figure 4: Calibration curve on OOT, 10 quantile bins. The systematic upward shift across the range is the expected behaviour of a TTC scorecard during a downturn.

4.3 Population stability

The Population Stability Index measures whether the grade distribution on a new cohort matches the training distribution:

$$\text{PSI} = \sum_g (\pi_g^{\text{new}} - \pi_g^{\text{train}}) \cdot \log \frac{\pi_g^{\text{new}}}{\pi_g^{\text{train}}}$$

Comparison	PSI
TRAIN vs VAL (2022)	0.0003
TRAIN vs OOT (2023)	0.0001

Table 5: PSI well below the conventional thresholds (0.10 stable, 0.25 material shift). The grade distribution barely moves across the five-year panel.

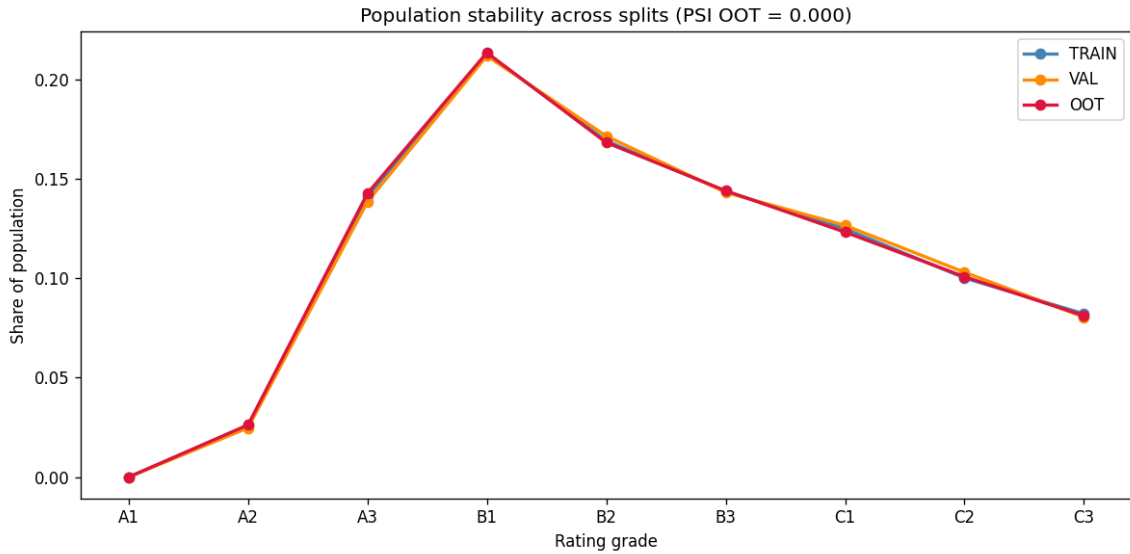


Figure 5: Share of population in each grade across train, validation, and OOT cohorts.

5 Regulatory framework

5.1 Basel II and III

Basel II (2004) introduced the Internal Ratings-Based (IRB) approach, which lets banks use their own PD, LGD, and EAD estimates to compute regulatory capital. The IRB capital formula for a single retail exposure depends most heavily on PD: a higher PD drives a higher risk weight, which drives a higher required capital reserve. Basel III (2010 onwards) tightened the capital requirements, added the leverage ratio and liquidity coverage ratio, and constrained how much banks could rely on internal models versus standardised risk weights.

5.2 IFRS 9

IFRS 9, effective from January 2018, replaced the incurred-loss model of IAS 39 with the Expected Credit Loss (ECL) framework for loan provisioning. Every loan is assigned to one of three stages:

- **Stage 1:** performing. ECL provision equals the expected loss over the next 12 months.
- **Stage 2:** significant increase in credit risk since origination. ECL provision equals the expected loss over the loan's remaining life.
- **Stage 3:** credit-impaired (default). Lifetime ECL on a defaulted basis.

The PD that drives staging and ECL under IFRS 9 differs from Basel in two ways. IFRS 9 uses a point-in-time (PIT) PD conditional on current macroeconomic conditions, and requires lifetime PD for Stage 2 and 3.

6 Compliance checklist

The tables below list the specific Basel and IFRS 9 requirements relevant to a retail PD scorecard, marked against what is present in this prototype and what is not.

6.1 Basel II/III requirements

Requirement	Status	In this project
Minimum 5 years of observation data	Met	2019–2023, ten half-year vintages
Master scale, ≥ 7 non-default grades	Met	9-grade A1–C3 scale
Through-the-cycle PD calibration	Met	Intercept-shifted to long-run-average
Chronological out-of-time validation	Met	2023 cohorts held out for backtest
Discrimination metrics reported	Met	Gini, KS, AUC per cohort
Stability monitoring on rating output	Met	PSI on grade distribution
Reasonable sample per grade	Met	Thousands of defaults per grade
12-month PD horizon as regulatory anchor	Partial	18-month performance window used
Defined default trigger (90+ DPD or UTP)	Not in scope	Binary flag, no DPD/UTP construct
Override capability	Not in scope	No manual override layer built
Independent validation	Not in scope	Prototype, no second-line review
Use test (real lending decisions)	Not in scope	Demonstration only
Model risk management framework	Not in scope	No MRM infrastructure
Supervisory approval (Finansinspektionen)	Not in scope	Portfolio project, not submitted

Table 6: Basel II/III compliance checklist. The structural and quantitative requirements relevant to model development are met. The governance and process requirements that a production deployment requires are out of scope for a portfolio prototype.

6.2 IFRS 9 requirements

Requirement	Status	In this project
Calibrated PD foundation suitable for ECL	Met	TTC-calibrated PD at grade level
Origination-grade infrastructure for SICR	Met	Master scale enables grade comparison
Stage 1/2/3 classification implementation	Not built	Infrastructure present, logic not coded
SICR triggers (PD ratio, days past due)	Not built	No SICR rule implemented
Lifetime PD projection	Not built	Single-horizon PD only
PIT PD via macro overlay	Not built	Only TTC version computed
Multi-scenario weighting (base/up/down)	Not built	Single deterministic scenario
Discounted lifetime ECL calculation	Not built	ECL formula not applied
LGD model	Not built	Out of scope for PD project
EAD model	Not built	Out of scope for PD project

Table 7: IFRS 9 compliance checklist. The PD model provides the foundation but the ECL engine, staging logic, macro overlay, and LGD/EAD models that complete IFRS 9 compliance are scheduled as separate projects in the portfolio.

6.3 Summary

This prototype covers the methodological core of a Basel IRB PD scorecard. The gaps are the governance and infrastructure layers that sit above the model (independent validation, MRM framework, use test, supervisory approval) and the IFRS 9 ECL engine that sits below it (staging, lifetime PD, macro overlay, LGD, EAD). Both gaps are expected for a portfolio project rather than a production deployment.

7 What the model shows

The scorecard discriminates reasonably (Gini ≈ 0.54 on OOT), is tightly calibrated per grade on the training data, and shows the expected mild under-prediction during the 2023 stress vintage. PSI

confirms that the grade distribution is stable across the full five-year panel.

The headline pattern is the OOT under-prediction. Observed default rates exceed predicted by 2–7 percentage points in the higher-risk grades. This is the textbook signature of a TTC-calibrated scorecard during a downturn: the model is anchored to a long-run-average that includes the benign pre-COVID years, and the 2023 cohort runs above that average. In production this would be absorbed by a PIT macro overlay for IFRS 9 ECL while keeping the TTC version for Basel regulatory capital.

8 Limitations

- Synthetic data. The bureau attributes are simulated, not from a real credit bureau. The IV pattern is plausible but not validated against external benchmarks.
- No defined default trigger. The target is a binary outcome flag without 90+ days past due or unlikely-to-pay construction.
- Single performance window. The 18-month outcome horizon is fixed across vintages.
- No monotone constraints on WoE. Some features end up with non-monotone WoE curves, which is statistically fine but a model governance committee would typically require monotonicity.
- No fairness audit. A production model would test for disparate impact across protected attributes.
- No champion-challenger framework. A production deployment would shadow the new model against the existing scorecard for 3–6 months before any traffic switch.
- Governance and infrastructure layers (independent validation, MRM framework, use test, supervisory approval) are out of scope.

9 References

- Basel Committee on Banking Supervision, *International Convergence of Capital Measurement and Capital Standards* (Basel II/III).
- European Banking Authority, *Guidelines on PD Estimation, LGD Estimation and the Treatment of Defaulted Exposures* (EBA/GL/2017/16).
- Siddiqi, *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring* (Wiley, 2012). Standard industry reference for the WoE/IV scorecard workflow.
- Anderson, *The Credit Scoring Toolkit* (Oxford, 2007).
- IFRS 9, *Financial Instruments*. The accounting standard that drives the staging engine and lifetime ECL.